Ejercicios de tcpdump

es una herramienta de **captura y análisis de tráfico de red** en modo texto (CLI). Sirve para **ver paquetes que pasan por una interfaz de red** en tiempo real (Ethernet/Wi-Fi), y es muy usada en administración de redes y ciberseguridad.

🔹 **Ejercicios básicos**

1. **Listar interfaces disponibles**

---

tcpdump -D

👉 Muestra todas las interfaces de red que tcpdump puede usar.

```
┌──(kali㉿kali)-[~]
└─$ tcpdump -D
1.eth0 [Up, Running, Connected]
2.eth1 [Up, Running, Connected]
3.any (Pseudo-device that captures on all interfaces) [Up, Running]
4.lo [Up, Running, Loopback]
5.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
6.nflog (Linux netfilter log (NFLOG) interface) [none]
7.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
8.dbus-system (D-Bus system bus) [none]
9.dbus-session (D-Bus session bus) [none]
```

2. **Capturar tráfico en la interfaz eth0**

---

tcpdump -i eth0

👉 Ver los primeros 20 paquetes que pasan por eth0.

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth1
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:29:04.591174 IP 192.168.0.108.61385 > 239.255.255.250.1900: UDP, length 170
11:29:04.593536 IP 192.168.0.29.61389 > 239.255.255.250.1900: UDP, length 170
11:29:04.618169 ARP, Request who-has 192.168.0.1 tell 192.168.0.53, length 28
11:29:04.620515 ARP, Reply 192.168.0.1 is-at 00:a0:26:d2:68:9a (oui Unknown), length 46
11:29:04.620524 IP 192.168.0.53.59857 > 254.red-80-58-61.staticip.rima-tde.net.domain: 15960+ PTR? 250.255.255.239.
in-addr.arpa. (46)
11:29:04.643156 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.59857: 15960 NXDomain 0/1/0 (103)
11:29:04.646520 IP 192.168.0.53.35547 > 254.red-80-58-61.staticip.rima-tde.net.domain: 41777+ PTR? 108.0.168.192.in
-addr.arpa. (44)
11:29:04.670242 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.35547: 41777 NXDomain 0/1/0 (121)
11:29:04.670520 IP 192.168.0.53.52596 > 254.red-80-58-61.staticip.rima-tde.net.domain: 9802+ PTR? 29.0.168.192.in-a
ddr.arpa. (43)
11:29:04.694487 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.52596: 9802 NXDomain 0/1/0 (120)
11:29:04.712573 IP 192.168.0.53.35147 > 254.red-80-58-61.staticip.rima-tde.net.domain: 10509+ PTR? 1.0.168.192.in-a
ddr.arpa. (42)
11:29:04.737048 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.35147: 10509 NXDomain 0/1/0 (119)
11:29:04.738096 IP 192.168.0.53.42994 > 254.red-80-58-61.staticip.rima-tde.net.domain: 59111+ PTR? 53.0.168.192.in-
addr.arpa. (43)
11:29:04.762930 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.42994: 59111 NXDomain 0/1/0 (120)
11:29:04.764199 IP 192.168.0.53.55043 > 254.red-80-58-61.staticip.rima-tde.net.domain: 56423+ PTR? 254.61.58.80.in-
addr.arpa. (43)
11:29:04.788148 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.55043: 56423 1/0/0 PTR 254.red-80-5
8-61.staticip.rima-tde.net. (95)
11:29:04.855643 IP 192.168.0.131.56488 > wb-in-f188.1e100.net.5228: Flags [.], seq 1955922806:1955922807, ack 18221
18130, win 253, length 1
11:29:04.891435 IP wb-in-f188.1e100.net.5228 > 192.168.0.131.56488: Flags [.], ack 1, win 1047, options [nop,nop,sa
ck 1 {0:1}], length 0
11:29:04.913270 IP 192.168.0.53.42650 > 254.red-80-58-61.staticip.rima-tde.net.domain: 45428+ PTR? 188.1.102.66.in-
addr.arpa. (43)
11:29:04.918914 IP 192.168.0.84.mdns > mdns.mcast.net.mdns: 0 ANY (QM)? DESKTOP-M7CM9BE._dosvc._tcp.local. (51)
11:29:04.918916 IP6 fe80::779a:eb5e:4232:4c83.mdns > ff02::fb.mdns: 0 ANY (QM)? DESKTOP-M7CM9BE._dosvc._tcp.local.
(51)
11:29:04.919766 IP 192.168.0.97.mdns > mdns.mcast.net.mdns: 0*- [0q] 1/0/3 SRV DESKTOP-M7CM9BE.local.:7680 0 0 (675
)
11:29:04.920218 IP 192.168.0.84.mdns > mdns.mcast.net.mdns: 0 ANY (QM)? DESKTOP-M7CM9BE(1)._dosvc._tcp.local. (54)
11:29:04.920218 IP6 fe80::d12d:c7e2:4ae3:1684.mdns > ff02::fb.mdns: 0*- [0q] 1/0/3 SRV DESKTOP-M7CM9BE.local.:7680
0 0 (675)
```

3. Guardar captura en un fichero

tcpdump -i eth0 -w trafico.log

👉 Luego ábrelo con:

tcpdump -r trafico.log

```
┌──(kali㉿kali)-[~]
└─$ tcpdump -r trafico.log
reading from file trafico.log, link-type EN10MB (Ethernet), snapshot length 262144
11:33:20.935182 ARP, Request who-has 192.168.0.101 (Broadcast) tell 192.168.0.90, length 46
11:33:20.957184 ARP, Request who-has 192.168.0.102 (Broadcast) tell 192.168.0.90, length 46
11:33:20.979972 ARP, Request who-has 192.168.0.103 (Broadcast) tell 192.168.0.90, length 46
11:33:21.006238 ARP, Request who-has 192.168.0.104 (Broadcast) tell 192.168.0.90, length 46
11:33:21.035921 ARP, Request who-has 192.168.0.105 (Broadcast) tell 192.168.0.90, length 46
11:33:21.062931 ARP, Request who-has 192.168.0.106 (Broadcast) tell 192.168.0.90, length 46
11:33:21.078723 ARP, Request who-has 192.168.0.107 (Broadcast) tell 192.168.0.90, length 46
11:33:21.097354 ARP, Request who-has 192.168.0.108 (Broadcast) tell 192.168.0.90, length 46
11:33:21.107158 ARP, Request who-has 192.168.1.196 tell 192.168.0.88, length 46
11:33:21.129300 ARP, Request who-has 192.168.0.109 (Broadcast) tell 192.168.0.90, length 46
11:33:21.148913 ARP, Request who-has 192.168.0.110 (Broadcast) tell 192.168.0.90, length 46
11:33:21.182022 ARP, Request who-has 192.168.0.111 (Broadcast) tell 192.168.0.90, length 46
11:33:21.196080 ARP, Request who-has 192.168.0.112 (Broadcast) tell 192.168.0.90, length 46
11:33:21.220926 IP 192.168.0.72 > 239.255.255.251: igmp v2 report 239.255.255.251
11:33:21.220927 IP 192.168.0.72 > mdns.mcast.net: igmp v2 report mdns.mcast.net
11:33:21.222923 ARP, Request who-has 192.168.0.113 (Broadcast) tell 192.168.0.90, length 46
11:33:21.271952 ARP, Request who-has 192.168.0.114 (Broadcast) tell 192.168.0.90, length 46
11:33:21.292447 ARP, Request who-has 192.168.0.115 (Broadcast) tell 192.168.0.90, length 46
11:33:21.324020 ARP, Request who-has 192.168.0.116 (Broadcast) tell 192.168.0.90, length 46
11:33:21.367497 ARP, Request who-has 192.168.0.117 (Broadcast) tell 192.168.0.90, length 46
11:33:21.406789 ARP, Request who-has 192.168.0.118 (Broadcast) tell 192.168.0.90, length 46
11:33:21.426792 ARP, Request who-has 192.168.0.119 (Broadcast) tell 192.168.0.90, length 46
11:33:21.473559 ARP, Request who-has 192.168.0.120 (Broadcast) tell 192.168.0.90, length 46
11:33:21.494686 ARP, Request who-has 192.168.0.121 (Broadcast) tell 192.168.0.90, length 46
11:33:21.562367 ARP, Request who-has 192.168.0.122 (Broadcast) tell 192.168.0.90, length 46
11:33:21.582344 ARP, Request who-has 192.168.0.123 (Broadcast) tell 192.168.0.90, length 46
11:33:21.603440 ARP, Request who-has 192.168.0.124 (Broadcast) tell 192.168.0.90, length 46
11:33:21.645858 ARP, Request who-has 192.168.0.125 (Broadcast) tell 192.168.0.90, length 46
11:33:21.673074 ARP, Request who-has 192.168.0.126 (Broadcast) tell 192.168.0.90, length 46
11:33:21.710170 IP 192.168.0.125.netbios-ns > 192.168.1.255.netbios-ns: UDP, length 50
11:33:21.751033 ARP, Request who-has 192.168.0.127 (Broadcast) tell 192.168.0.90, length 46
11:33:21.774533 ARP, Request who-has 192.168.0.128 (Broadcast) tell 192.168.0.90, length 46
11:33:21.862569 IP 192.168.0.109.54642 > 239.255.255.250.1900: UDP, length 170
```

- ## Filtrado por protocolos y puertos

4. Capturar solo tráfico TCP en eth0

tcpdump -i eth0 tcp

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth1 tcp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:36:55.896291 IP 192.168.0.131.62649 > 254.red-80-58-61.staticip.rima-tde.net.domain: Flags [S], seq 3673733553,
win 65535, options [mss 1460,nop,wscale 8,nop,nop,sackOK], length 0
11:36:55.896292 IP 192.168.0.131.54582 > 254.red-80-58-61.staticip.rima-tde.net.domain: Flags [S], seq 3563271960,
win 65535, options [mss 1460,nop,wscale 8,nop,nop,sackOK], length 0
11:36:55.918506 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.131.54582: Flags [S.], seq 1582799878,
 ack 3563271961, win 28960, options [mss 1460], length 0
11:36:55.918758 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.131.62649: Flags [S.], seq 2253205254,
 ack 3673733554, win 28960, options [mss 1460], length 0
11:36:55.918759 IP 192.168.0.131.54582 > 254.red-80-58-61.staticip.rima-tde.net.domain: Flags [.], ack 1, win 65535
, length 0
11:36:55.919002 IP 192.168.0.131.62649 > 254.red-80-58-61.staticip.rima-tde.net.domain: Flags [.], ack 1, win 65535
, length 0
11:36:55.919003 IP 192.168.0.131.62649 > 254.red-80-58-61.staticip.rima-tde.net.domain: Flags [P.], seq 1:3, ack 1,
 win 65535, length 2 [prefix length(33) ≠ length(0)] (invalid)
11:36:55.919180 IP 192.168.0.131.62649 > 254.red-80-58-61.staticip.rima-tde.net.domain: Flags [P.], seq 3:36, ack 1
, win 65535, length 33 [prefix length(42403) ≠ length(31)] (invalid)
11:36:55.919180 IP 192.168.0.131.54582 > 254.red-80-58-61.staticip.rima-tde.net.domain: Flags [P.], seq 1:3, ack 1,
 win 65535, length 2 [prefix length(33) ≠ length(0)] (invalid)
11:36:55.919181 IP 192.168.0.131.54582 > 254.red-80-58-61.staticip.rima-tde.net.domain: Flags [P.], seq 3:36, ack 1
, win 65535, length 33 [prefix length(31043) ≠ length(31)] (invalid)
11:36:55.922546 IP 192.168.0.131.56744 > 250.red-80-58-61.staticip.rima-tde.net.domain: Flags [S], seq 1713412218,
win 65535, options [mss 1460,nop,wscale 8,nop,nop,sackOK], length 0
11:36:55.922908 IP 192.168.0.131.59225 > 250.red-80-58-61.staticip.rima-tde.net.domain: Flags [S], seq 2560311685,
win 65535, options [mss 1460,nop,wscale 8,nop,nop,sackOK], length 0
11:36:55.941393 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.131.54582: Flags [.], ack 3, win 29200
, length 0
11:36:55.941394 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.131.54582: Flags [.], ack 36, win 2920
0, length 0
11:36:55.941395 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.131.54582: Flags [P.], seq 1:52, ack 3
6, win 29200, length 51 31043 1/0/0 A 142.250.200.131 (49)
11:36:55.941395 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.131.62649: Flags [.], ack 3, win 29200
, length 0
```

5. **Capturar tráfico en el puerto 80 (HTTP)**

tcpdump -i eth0 tcp port 80

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 tcp port 80
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

6. **Capturar tráfico DNS (UDP puerto 53)**

tcpdump -i eth0 udp port 53

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 udp port 53
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

- **Filtrado por IP**

7. **Tráfico desde una IP concreta (host origen)**

tcpdump -i eth0 src 192.168.43.143

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 src 192.168.0.53
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

8. **Tráfico hacia una IP concreta (host destino)**

tcpdump -i eth0 dst 192.168.43.143

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 dst 192.168.0.53
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

9. **Tráfico de y hacia una IP concreta (host completo)**

tcpdump host 192.168.1.143

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump host 192.168.0.113
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

- **Filtrado por MAC y red**

10. Tráfico con destino a una dirección MAC

tcpdump ether dst 8A:B1:11:A0:BC:53

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump ether dst 8:A:B1:11:A0:BC:53
tcpdump: bogus ethernet address 8:A:B1:11:A0:BC:
```

11. Tráfico de una red completa

tcpdump net 192.168.43.0/24

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump net 192.168.0.0/24
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

- ### Visualización de datos

12. Mostrar el contenido en ASCII

tcpdump -i eth0 -A

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth1 -A
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:00:36.343764 IP 192.168.0.131.7680 > 192.168.0.46.60704: Flags [P.], seq 1061335793:1061335797, ack 2855592116,
win 255, length 4
E..,x.@............... ?B...4..P............
12:00:36.395986 ARP, Request who-has 192.168.0.124 tell 192.168.0.122, length 46
.........\..l...z.........|..................
12:00:36.397418 IP 192.168.0.46.60704 > 192.168.0.131.7680: Flags [.], ack 4, win 255, length 0
E..(!B@...W..........  ...4..?B..P............
12:00:36.406791 IP 192.168.0.53.49437 > 254.red-80-58-61.staticip.rima-tde.net.domain: 22858+ PTR? 46.0.168.192.in-
addr.arpa. (43)
E..G..@.@.>....5P:=....5.3OZYJ..........46.0.168.192.in-addr.arpa.....
12:00:36.430758 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.49437: 22858 NXDomain 0/1/0 (120)
E.....@.4..HP:=....5.5.....0YJ..........46.0.168.192.in-addr.arpa.............+.A.prisoner.iana.org.
hostmaster.root-servers.EwT...........  :..
12:00:36.431346 IP 192.168.0.53.34282 > 254.red-80-58-61.staticip.rima-tde.net.domain: 52469+ PTR? 131.0.168.192.in
-addr.arpa. (44)
E..H..@.@......5P:=....5.4O[.............131.0.168.192.in-addr.arpa.....
```

13. Mostrar el contenido en hexadecimal

tcpdump -i eth0 -XX

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth1 -A
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:00:36.343764 IP 192.168.0.131.7680 > 192.168.0.46.60704: Flags [P.], seq 1061335793:1061335797, ack 2855592116,
win 255, length 4
E..,x.@................ ?B...4..P.............
12:00:36.395986 ARP, Request who-has 192.168.0.124 tell 192.168.0.122, length 46
..........\..l...z.........|..................
12:00:36.397418 IP 192.168.0.46.60704 > 192.168.0.131.7680: Flags [.], ack 4, win 255, length 0
E..(!B@...W..........  ...4..?B..P.............
12:00:36.406791 IP 192.168.0.53.49437 > 254.red-80-58-61.staticip.rima-tde.net.domain: 22858+ PTR? 46.0.168.192.in-
addr.arpa. (43)
E..G..@.@.>....5P:=....5.3OZYJ...........46.0.168.192.in-addr.arpa.....
12:00:36.430758 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.49437: 22858 NXDomain 0/1/0 (120)
E.....@.4..HP:=....5.5.....0YJ...........46.0.168.192.in-addr.arpa..............+.A.prisoner.iana.org.
hostmaster.root-servers.EwT...........  :..      :.
12:00:36.431346 IP 192.168.0.53.34282 > 254.red-80-58-61.staticip.rima-tde.net.domain: 52469+ PTR? 131.0.168.192.in
-addr.arpa. (44)
E..H..@.@......5P:=....5.4O[.............131.0.168.192.in-addr.arpa.....
12:00:36.455097 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.34282: 52469 NXDomain 0/1/0 (121)
E....δ@.4.
.P:=....5.5....R>.............131.0.168.192.in-addr.arpa................A.prisoner.iana.org.
hostmaster.root-servers.FwT...........  :..      :.
12:00:36.456068 IP 192.168.0.53.40082 > 254.red-80-58-61.staticip.rima-tde.net.domain: 30045+ PTR? 124.0.168.192.in
-addr.arpa. (44)
E..HD[@.@..4...5P:=....5.4O[u]...........124.0.168.192.in-addr.arpa.....
12:00:36.479818 IP 254.red-80-58-61.staticip.rima-tde.net.domain > 192.168.0.53.40082: 30045 NXDomain 0/1/0 (121)
E.....@.5._4P:=....5.5......u]...........124.0.168.192.in-addr.arpa................A.prisoner.iana.org.
hostmaster.root-servers.FwT...........  :..      :.
12:00:36.480120 IP 192.168.0.53.42741 > 254.red-80-58-61.staticip.rima-tde.net.domain: 58378+ PTR? 122.0.168.192.in
-addr.arpa. (44)
E..Hq.@.@.z....5P:=....5.4O[.
...........122.0.168.192.in-addr.arpa.....
12:00:36.487429 IP 192.168.0.43.mdns > mdns.mcast.net.mdns: 0*- [0q] 1/0/6 PTR Aula 3._airplay._tcp.local. (604)
```

🔹 Ejercicios combinados (operadores lógicos)

14. **Tráfico desde una IP y en puerto 80**

tcpdump -i eth0 src 192.168.43.143 and tcp port 80

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth1 src 192.168.0.53 and tcp port 80
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

15. **Tráfico desde una IP excluyendo UDP**

tcpdump -i eth0 src 192.168.1.143 and not udp

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth1 src 192.168.0.53 and not udp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C
```

16. **Tráfico desde una IP en HTTP o HTTPS**

tcpdump -i eth0 src 192.168.1.143 and (port http or https)

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth1 src 192.168.0.53 and (port http or https)
zsh: unknown sort specifier
```

👉 Con esta lista de **16 ejercicios** tienes un recorrido completo:

- Empezando por capturar y guardar tráfico.
- Luego filtrando por protocolo, puerto, IP, MAC y red.
- Finalmente usando operadores lógicos y visualización en distintos formatos.